

Memory in pieces

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Remembering requires a description that preserves the distinctions an enquiry can make. A coarse description succeeds only if its retained observables and their effective law survive the next change of scale. We give a finite construction, Parallax, in which each local object carries its mass, internal modes, decoder, source incidence and unresolved contribution. Spectral selection supplies the internal modes directly. Exact composition identities make the same operation repeatable, while elimination of discarded modes produces a memory kernel. Compatible semantic observations form a joint source-labelled state whose conditional continuations admit compact tensor representations. On a 4 GB laptop GPU, exact Boolean queries over 262,144 distinct correlated states take about 60 microseconds per batch of 1024, against 1491 microseconds for direct evaluation. A separate news experiment answers 40 of 48 questions, against 39 for flat RAG and 35 for RAPTOR. A metadata control also scores 40 with identical contexts; neither registered paired difference passes the corrected significance threshold. On the new fixed sets, the witness reader nearly ties dense QASPER F1 and answers 30/48 LongMem questions versus dense 33/48. These experiments identify the remaining interface problem: numerical fidelity to an embedding does not by itself preserve answer-bearing evidence. The theory specifies what must travel between scales; the experiments locate where the present language representation still discards it.

I WHAT MUST A MEMORY PRESERVE?

Suppose a record contains three binary observations, A, B, C , and allows only the four equally likely configurations 000, 011, 101, 110. Every single observation is unbiased. Every pair is independent. Yet knowing $A = B = 0$ determines $C = 0$. A memory containing only the separate field summaries cannot answer this question. A memory retaining the joint law can.

This small example contains the central difficulty of corpus reconstruction. A document has many distinguishable aspects: who acts, what happens, when it happens, which conditions hold and what follows. Their separate descriptions are useful, but a question often concerns their relation. A sentence may establish an event; a paragraph its cause; a chapter the circumstances that make the cause intelligible. Scale is therefore tied to the enquiry. Text length alone does not determine which description is adequate.

The physical problem is familiar. Eliminating degrees of freedom changes both the observables and the law governing those that remain. An effective theory must retain the relevant distinctions, account for their interactions and provide a way to resolve the neglected part. Here the state is a classical distribution over recorded observations. The Hilbert-space and tensor constructions below are representations of that state and its tests; all computation is classical.

The Synthetics programme develops this requirement through adaptive normal forms, operatorial loss, self-consistent loss geometry and statistical lawhood [1–4]. Its recurring lesson is that a description and the criterion used to judge it must transform together. Repeatedly compressing the output of the previous compression is inadequate when the previous output omitted the very register needed to assess its loss. That mechanism also

explains how recursive synthesis can preserve familiar averages while progressively erasing rare distinctions.

We use four terms with specific roles. A *semantic axis* is a declared family of observations, such as event time or mechanism. A *section* is a local source support on which a reference measure, observations and comparison rule are defined. A *fibre* is the family of admissible representations of those local observations; a frame chooses coordinates within it. A *grade* is an accepted effective description, specified by which tests and operations it preserves to declared tolerances. Neither rank nor depth alone specifies a grade.

Parallax constructs these objects during ingestion and selects among them during reconstruction. Its recursive operation, the *fibred upfold*, passes a child’s complete retained object to its parent: positive mass, internal coordinates, effective law, decoder, source ownership and residual. Different axes can have different local hierarchies. Their common source incidence joins compatible observations without forcing them into one global semantic frame. Three questions organise the argument: which information determines the enquiry, how that information survives a change of grade, and when sharing its representation reduces the work. We first derive the operation, then show which parts of it have been verified mathematically, which yield a computational gain, and what the evidence experiments require of the language interface.

II AN OBJECT WITH INTERNAL MODES

Fix a finite context with source states $i = 1, \dots, n$ and reference law $\pi_i > 0$, normalised to one. States are column probability vectors. Let C aggregate a partition of the sources and put $W = \text{diag}(\pi)$ and $W_c = \text{diag}(C\pi)$.

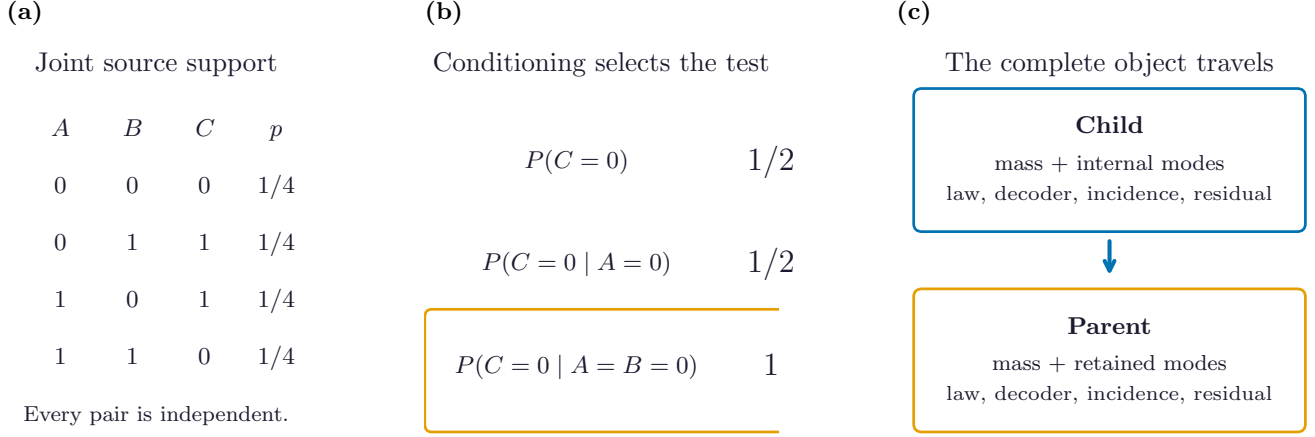


FIG. 1. The information an enquiry needs. (a) Four possible source records have uniform single and pair marginals. (b) Conditioning on two observations nevertheless determines the third. Separate field summaries lose this relation. (c) At a change of grade, the child's retained distinctions and the rule used to assess them enter the parent together. A residual certificate directs refinement; the retained source and decoder supply the omitted detail.

The conditional lift

$$R = WC^T W_c^{-1}, \quad CR = I \quad (1)$$

restores the reference conditional distribution inside each cell. It gives the exact decomposition $p = RCp + (I - RC)p$. The second term records which distinctions were removed.

Mass alone can close perfectly while erasing the enquiry. Consider a test whose values on four equally weighted sources are $(0, 1, 0, 1)$. Grouping the first pair and the second pair gives two means of $1/2$. If each child also retains its internal contrast, the test is exact within the children. Passing only their means to the next grade loses a root-mean-square test amplitude of $1/2$. An error account based on the children's retained contrasts would nevertheless report zero. The parent must receive the whole retained object.

To retain a distinction, we need coordinates for departures from the reference distribution. The appropriate linear coordinates follow from that measure. Define the whitened fluctuation and a reversible generator by

$$y = W^{-1/2}(p - \pi), \quad L = W^{-1/2}GW^{1/2}, \quad (2)$$

Here G is a declared generator on the source observations: its columns specify admitted transitions. In the controlled physical examples it is the given dynamics. In the news implementation it is a reversible diffusion constructed from the observed field vectors, as specified in App. H. The choice of this law defines which dynamics the grade must preserve. We require $G_{ij} \geq 0$ for $i \neq j$, $\mathbf{1}^T G = 0$ and $G_{ij}\pi_j = G_{ji}\pi_i$. Then $L = L^T \leq 0$ and $\|y\|^2 = \chi^2(p|\pi)$.

The partition supplies orthonormal constant columns $U_0 = W^{1/2}C^T W_c^{-1/2}$. Add block-supported columns B ,

orthogonal to those constants, and form an isometry $U = [U_0, B]$. The retained coordinates and operator are

$$z = U^T y, \quad A = U^T L U, \quad P = U U^T. \quad (3)$$

The mass coordinates obey $z_0 = W_c^{-1/2}(Cp - \pi_c)$, so they are determined by the separately stored positive coarse mass Cp . The additional coordinates $z_B = B^T y$ are the signed internal register. Their decoder, reference, operator, incidence and unresolved contribution travel with them. The residual certificate bounds the lost effect; it does not contain the omitted source values. Those remain reachable through the original source register and its refinement maps. This distinction is essential to executable reconstruction. The compact implementation stores task-wise squared losses and blockwise bounds on the dynamical defect, alongside the retained maps and source ownership.

For a source observable f , let $a = W^{1/2}f$. The decoder evaluates $f^T \pi + (U^T a)^T z$. Its difference from $f^T p$ is controlled by the residual $e_f = (I - P)a$. Since the stationary mass direction is retained,

$$|\delta\langle f \rangle_p| \leq \|e_f\| \sqrt{\chi^2(p|\pi)}. \quad (4)$$

This identifies both the lost distinction and the measure under which its size was assessed. For the conditional reference state $p = \pi(\cdot | \mathcal{E})$, where the event has reference mass $b = \pi(\mathcal{E}) > 0$, the multiplier is $\sqrt{b^{-1} - 1}$. Rare enquiries can therefore resolve directions that were small in the reference average.

III GRADING THE WHOLE OBJECT

The generator supplies a second preservation requirement. Its unresolved coupling is

$$D = U^\top L - AU^\top = U^\top L(I - P). \quad (5)$$

Both L and A generate contractions in the whitened Euclidean norm, so

$$\|U^\top e^{tL} - e^{tA}U^\top\|_2 \leq t\|D\|_2. \quad (6)$$

The right-hand side bounds the error per unit whitened state excursion. Observable preservation and dynamical closure are checked together. Keeping only total mass gives a closed generator for any Markov system, which is why closure alone cannot choose an adequate grade.

The required internal coordinates come from a direct spectral problem. A task column records the value of one observable on every source. A law-demand column records how the proposed coarse law couples to an omitted direction. Within each proposed parent, project the declared task columns and law-boundary columns orthogonal to its mass constant. Collect their weighted demands in a matrix E . Among local rank- r orthogonal projectors, the leading eigenspace of $EE^\top$ minimises the squared omitted demand. An equivalent SVD avoids forming that square matrix. The reference measure, task weights and relative law weight define this spectrum. The selected modes are then subjected to the actual task and closure tests, because retaining a mode can reveal another dynamical coupling.

Diffusion spectra propose neighbouring objects that might form a useful parent. Acceptance comes from the retained tests and operator. Thus spectral clustering and spectral correction have different jobs: one proposes a section, and the other supplies its internal degrees of freedom. Rank is a resource within that section. Grade specifies which effective description has been accepted.

A second grade acts on the same kind of object. Extend the proposed grouping to every coordinate owned by each child, including its internal modes. This is the fibred upfold in explicit coordinates. With isometries U_1, U_2 , transport

$$L_2 = U_2^\top L_1 U_2, \quad T_2 = U_2^\top T_1, \quad V_2 = U_2^\top V_1. \quad (7)$$

Here T contains whitened task columns and V the field decoder. The composite map is $U_{02} = U_1 U_2$. Its dynamical defect obeys

$$D_{02} = U_2^\top D_{01} + D_{12} U_1^\top, \quad (8)$$

while the squared task residuals add orthogonally. These identities make the repeated operation testable. They also show precisely why dropping a child's internal register invalidates an inherited error account.

The physical interpretation is particularly simple. The positive operator $H = -L$ is the Dirichlet, or dissipation, operator of the reversible law, and $-A = U^\top H U$ is its

Ritz compression. Eliminating the unresolved fluctuation $\eta = (I - P)y$ gives the exact equation

$$\begin{aligned} \dot{z}(t) &= Az(t) + D e^{tL_Q} \eta(0) \\ &+ \int_0^t D e^{(t-s)L_Q} D^\top z(s) ds, \end{aligned} \quad (9)$$

where $L_Q = (I - P)L(I - P)$ acts on the unresolved subspace. This is the finite linear projection-operator construction [5]. The same coupling measured by D generates memory in the reduced dynamics. When $D = 0$, the retained dynamics closes exactly. Keeping the modes that carry this coupling provides a controlled route to a more adequate effective object.

IV THE QUERY SELECTS A SECTION

Observations are compatible here when they can be jointly assigned to the same source under one declared reference law. Object, time and cause fields on a source are one example. Different operational contexts without such a joint assignment retain separate descriptions. Compatible observations admit a tensor description without requiring one semantic basis for every enquiry. The displayed tensor formula assumes that every selected field has an observation on the declared source support. Missing observations remain in an explicit unknown section and receive no fabricated field value. Suppose source i has unit observed vectors v_{if} in fields f . The augmented unit vector $a_{if} = (1, v_{if})/\sqrt{2}$ defines

$$\tau_i = \bigotimes_f |a_{if}\rangle \langle a_{if}|, \quad \Omega = \sum_i \pi_i |i\rangle \langle i| \otimes \tau_i. \quad (10)$$

The classical source register retains the correlations and provenance between fields. A query with unit vectors q_f chooses rank-one effects on its selected fields and identities on the others. Its source-conditioned score is

$$f_q(i) = \prod_{f \in S} \left(\frac{1 + \langle v_{if}, q_f \rangle}{2} \right)^2. \quad (11)$$

The product is evaluated for the same source. Separately averaging each factor would erase the coupling needed by a joint enquiry.

The positive map $\Phi(p) = \sum_i p_i \tau_i$ connects this description to the source-state calculus: every effect defines an ordinary source test. Its weighted column can be transported through Eq. (7). The feature density operator, the source fluctuation and a categorical amplitude state have different domains and spectra. Their decoder maps provide the connection.

The representations have distinct experimental roles. Signed source coordinates implement the recursively transported effective law. Categorical continuations implement exact joint counts and the parity scaling experiment. Local vector frames bound embedding scores,

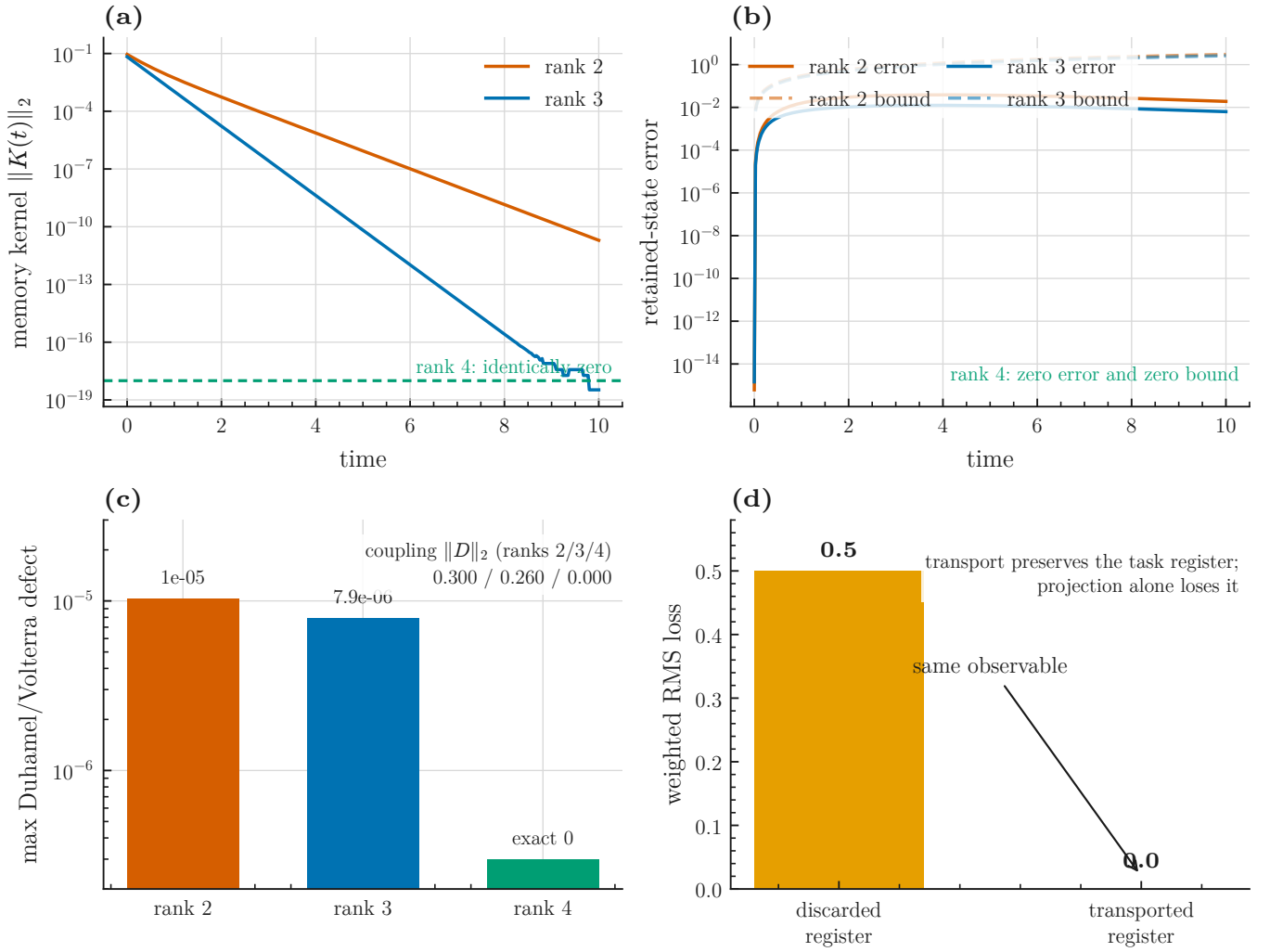


FIG. 2. A finite physical example of retained and eliminated modes. The resolved dynamics acquires the exact memory kernel generated by its coupling to the unresolved space. Increasing the retained space reduces that coupling. The accompanying observable example shows why mass alone cannot inherit a child’s accuracy: carrying the internal register changes a reference RMS loss of 1/2 to zero. All curves come from the saved four-state calculation; any quadrature error belongs to the displayed numerical integration.

while the language reader receives original passages. Their numerical costs and semantic guarantees belong to those respective interfaces.

Two further representations make the query operation concrete. A categorical tensor network contracts a compatible joint law, while the real-text benchmark uses local vector frames and original passages. Their error contracts connect through declared tests, and neither representation requires the other as a computational stage.

The parity example makes the distinction between ranks concrete. The diagonal distribution on d even-parity bits has 2^{d-1} nonzero entries, while its conditional continuation needs only two states: even and odd running parity. Its square-root amplitude also has Schmidt rank two across every nontrivial cut. The large number of worlds is therefore compatible with a small exact continuation. A generic high-entropy law need not have this

property.

The matrix-product-state connection applies directly to a compatible categorical law $p(x_1, \dots, x_d)$. An exact conditional-continuation representation gives an amplitude $\psi(x) = \sqrt{p(x)}$ as a product of local tensors. Canonicalisation exposes the Schmidt spectrum seen through the surrounding factors, so a change of query or conditioning changes the relevant environment [6]. Direct QR and SVD operations supply the representation and its truncations. No model weights are trained. The final contracted fidelity, together with the conditioning mass, gives an answer-error certificate, as derived in App. F.

For vector observations, take an orthonormal frame $B_b^\top B_b = I$ and coordinates $c_j = B_b^\top (v_j - \mu_b)$. Write

$$v_j = \mu_b + B_b c_j + r_j, \quad B_b^\top r_j = 0. \quad (12)$$

The query projection onto the unresolved directions

bounds the missing contribution:

$$|\langle r_j, q \rangle| \leq \|r_j\| \|(I - B_b B_b^\top)q\|. \quad (13)$$

This supplies member score intervals. A top- k reconstruction may stop when every omitted upper bound lies below the k th accepted lower bound, within the declared tolerance. Ambiguous members require a finer description or their original observations. Returned candidates are ordered by their resolved scores, because ordering determines which evidence fits a finite reader context.

This rule also fixes the passage boundary. A short article interpretation can propose useful evidence, but excluding an article from a passage enquiry requires a bound on its admitted passages. In the final news readout, exact source and date conditions select the supported source section, and the fine target ranks original passages. The article description remains available without becoming an unsupported hard gate.

V EXACT CONTINUATION ON THE GPU

A continuation state represents the conditional law of the factors still to come. Two prefixes share a state when they induce the same conditional suffix law. This identifies repeated structure in the joint distribution, rather than replacing the factors by independent marginals. A product query places an effect on each selected factor and the identity on the others. Messages contract through the shared continuation once per query.

Integer empirical weights admit an especially efficient Boolean lane. Each state stores a primitive denominator T_s and integer edge multipliers K_{sa} . For the allowed-symbol indicator $q_a \in \{0, 1\}$, the message is

$$m_s(q) = \sum_a K_{sa} q_a m_{b(a)}(q), \quad 0 \leq m_s \leq T_s. \quad (14)$$

Checked denominators make the result exact and determine the safe arithmetic width before execution. App. K derives the multipliers. Query cost depends on the visited continuation and batch size. The source count enters the construction cost and the irreducible width of the law.

The registered matrix contains 524 cells. The exact kernel executes in 500; 24 high-entropy cells exceed its declared width capacity and are refused. All executed exact answers agree with direct evaluation. The principal family has $N = 4096$ to 262 144 distinct even-parity configurations, with query support ranging from the whole law to one configuration. Its continuation width remains two. At the largest size, the median full API time over the query regimes is $59.8 \mu\text{s}$ for a batch of 1024, versus $1490.7 \mu\text{s}$ for a packed direct scan and $1739.6 \mu\text{s}$ for a weighted unique-row scan. A single query takes $37.4 \mu\text{s}$ versus $45.8 \mu\text{s}$ for the packed scan. Batching exposes the largest saving.

The largest continuation and integer add-on occupy about 3.4 kB, in addition to the source and witness data.

Extra construction costs about 55 ms. Warm, precompiled amortisation requires roughly 6500 single queries or 29 000–41 000 batched queries, depending on batch size. First-use JIT is recorded separately. On a repeated 64-world support, the weighted direct control is faster than recursion. These controls establish the useful regime: repeated conditional structure makes the representation economical.

The retained-mode operator construction has an independent verification. Across twelve grades of the 609-source news corpus, the compact GPU and dense reference operators differ by at most 2.32×10^{-13} , and transported tasks by 5.30×10^{-14} . A fixed-width sparse family scales linearly in stored operator entries and accumulation work up to 65,536 sources. The full child register is retained at every step.

An exact vector-search controller was also tested. Local PCA frames and residual-radius bounds return correct scores, but the tested binary tree opens too much of the corpus. Its raw-read scaling on clustered synthetic data is approximately $N^{1.10}$ – $N^{1.18}$, and it reads every source in the QASPER prefixes. Direct GPU matrix multiplication is faster. The continuation result and this geometric result concern different representations of the query and cannot be exchanged.

VI FROM A NUMERICAL OBSERVATION TO EVIDENCE

A cosine score is a well-defined numerical observable. Whether the associated passage answers a question is a further claim. The distinction becomes visible when a selector accurately reconstructs the question embedding but omits a passage containing the requested fact. The geometric residual can shrink while evidence recovery gets worse.

The language readout therefore keeps explicit evidence obligations and source bindings. Exact observed meta-data can restrict support. Each unresolved semantic obligation receives candidate witnesses from its own score stream. Compatible witnesses must share the required source or entity binding. Original passages, their identifiers and any required adjacent context are packed under the same token budget as the controls. Unknown bindings remain unknown. The fixed reader then evaluates what the supplied text actually supports.

This interface is deliberately visible in the experiments. A question about how many projects were completed needs an aggregation over related events; a highly similar passage about one project is insufficient. A comparison needs both sides, with their qualifications intact. Query projection and source diversity are not replacements for those operations.

The public news comparison uses MultiHop-RAG: 609 articles, non-overlapping 100-token passages and 48 fixed questions. Every arm uses the same BGE encoder, Sol reader and 2000-token context budget. Source-

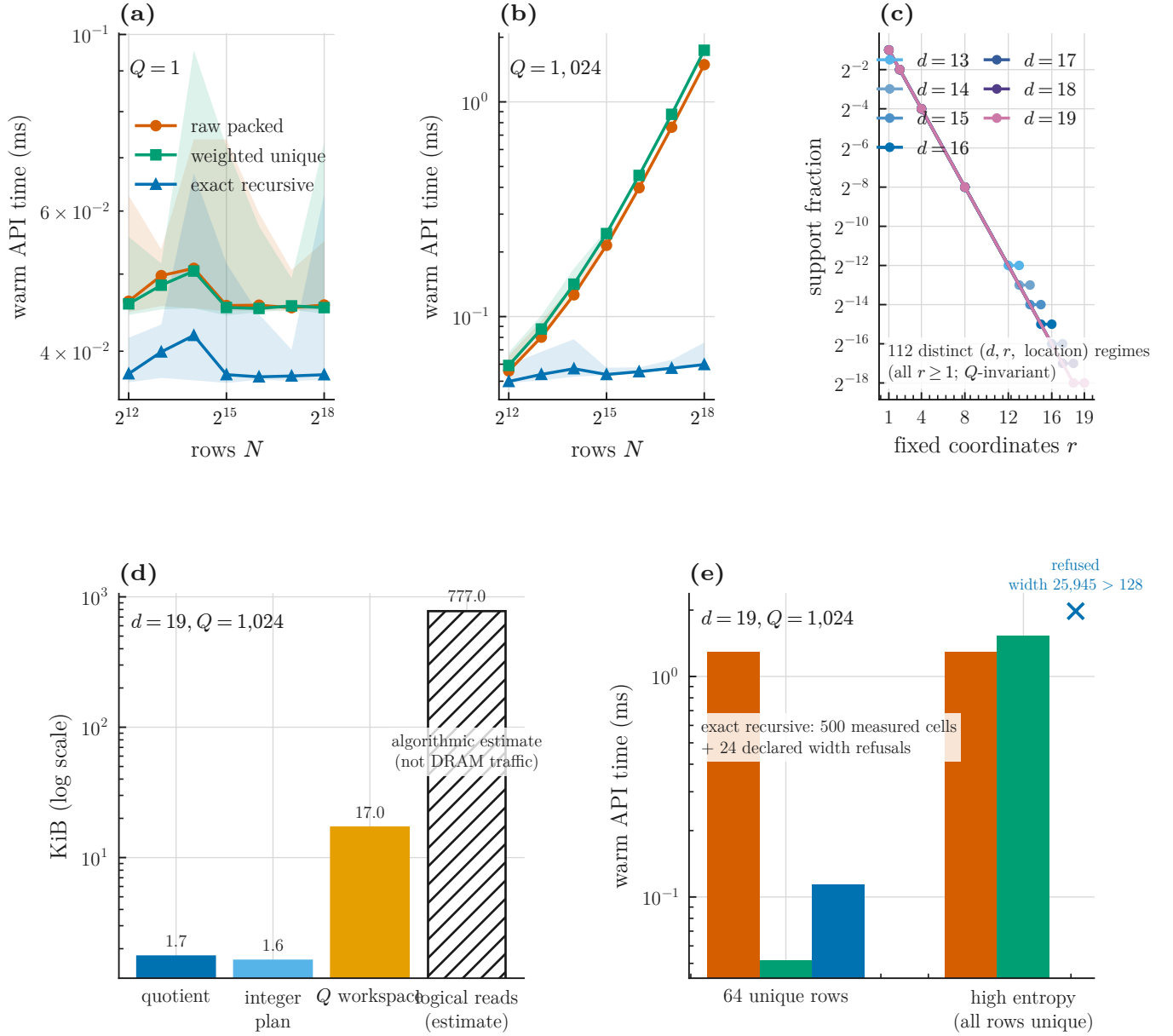


FIG. 3. Exact queries on the unique even-parity family. The continuation retains correlations while reusing a width-two conditional law. Source size, query support and batch size are shown separately so that locality is not confused with corpus size. Bytes counted by the algorithm are logical loads, not measured device-memory traffic. The input law and witness storage remain separate from the compact continuation. Construction and cold compilation are reported with the run rather than absorbed into warm query latency.

conditioned retrieval answers 40 correctly, flat RAG 39 and independent RAPTOR 35. A flat metadata control also answers 40, with byte-identical contexts. The registered paired Holm probabilities are 0.5 against RAG and 0.0625 against RAPTOR. Thus the observed advantage is a descriptive result on this sample. App. I records both disjoint trials, preprocessing and the complete protocol.

The extension development tests use 24 LongMemEval [7] cases and 32 QASPER [8] training questions. Of these, 18 and 31 respectively have an eligible positive

evidence target. At 4000 tokens, the witness readout supplies every annotated turn on 15/18 LongMem cases, versus 16/18 for dense and 15/18 for hybrid retrieval. On QASPER it supplies complete evidence on 17/31, tying dense and exceeding hybrid's 15/31. Only 24 of the 31 QASPER questions have one annotation whose required evidence is fully mapped to the source bank; unmapped requirements remain in the denominator.

The full-question-plus-child experiment does not improve the four affected LongMem cases. Its failure audit

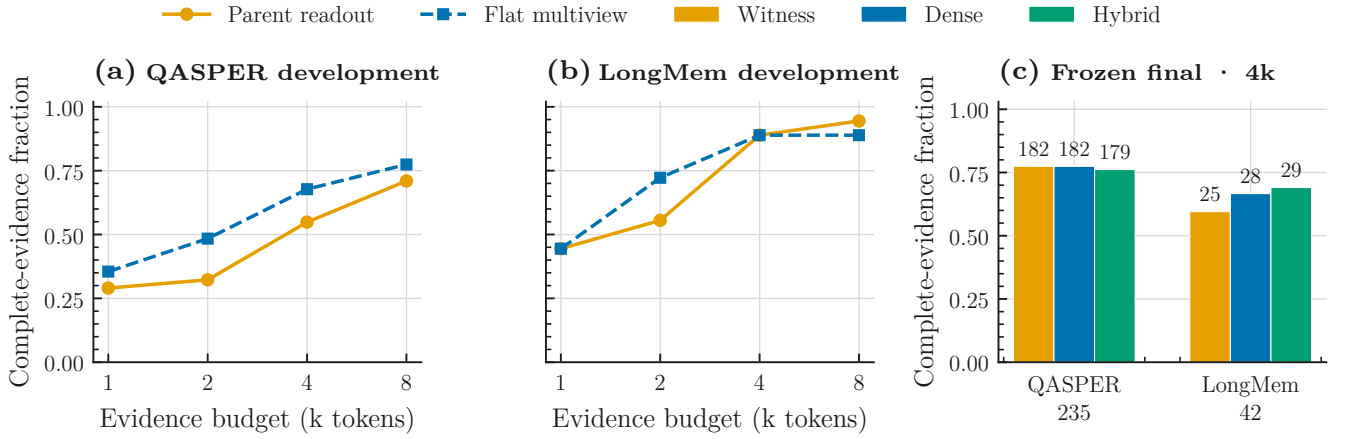


FIG. 4. Evidence recovery across budgets and datasets. (a,b) Development comparisons retain original parent passages while adding local span views. Maximum pooling changes both granularity and the number of scoring opportunities, so this is a combined development diagnostic. (c) The frozen final witness method and its controls at 4000 tokens, on 235 eligible QASPER questions and 42 LongMem questions. Every point measures complete supplied evidence; reader answer accuracy is evaluated separately.

separates a missing relational observation from a packing effect: first-fit packing can admit a long earlier passage at a larger budget and thereby displace a later necessary passage. Accordingly, coverage need not be monotone under that implemented packing rule.

The final fixed QASPER set contains 256 questions, of which 235 have eligible positive evidence annotations. Under the 4000-token budget, both the frozen witness method and dense retrieval recover complete evidence on 182/235 (77.45%), versus 179/235 for hybrid and 171/235 for lexical retrieval. The witness and dense contexts are identical on 252 of 256 questions. Median scoring, selection, transfer and packing times are 3.11 ms for witness, 2.75 ms for dense and 3.95 ms for hybrid. The complete five-arm, four-budget matrix was sealed before evidence scoring.

A final development-only source-view experiment retains each original parent passage and adds its exact constituent spans. The parent score is the maximum over these views. At 4000 tokens, QASPER complete evidence rises from 17/31 to 21/31, while LongMem’s flat readout stays at 16/18. At 2000 tokens the corresponding changes are 10/31 to 15/31 and 10/18 to 13/18. Parent passages remain the supplied evidence units. More finely split parents also receive more scoring opportunities, so the experiment measures the combined effect of local views and that multiplicity. It is separate from the frozen final method and supplies a bounded direction for later representation work.

The final LongMem sample contains 48 questions, including six abstentions. Of the 42 questions with positive reference evidence, complete 4000-token coverage is 25/42 for witness, 28/42 for dense and 29/42 for hybrid. These final retrieval results fix the supplied contexts for the shared reader trial. They identify a remaining loss in

TABLE I. The fixed shared-reader trial. QASPER reports native answer token F1 on 32 questions. LongMem reports correct answers under the official per-type rubric applied by the predeclared Sol judge on 48 questions.

Readout	QASPER F1	LongMem correct
Witness	0.46354	30/48
Dense	0.46396	33/48
Hybrid	0.44368	34/48

the witness interface on this sample, with no change to the method after final scoring.

The shared reader trial contains 80 fixed questions and three arms. Exact request sharing reduces the 240 matrix rows to 164 reader calls. Every reader and judge record succeeds. Table I gives the resulting answers. QASPER witness minus dense F1 is -0.00042 , with descriptive paired 95% bootstrap interval $[-0.00126, 0]$; against hybrid it is 0.01986 , with interval $[-0.00397, 0.04612]$. LongMem witness minus dense correctness is -6.25 percentage points, and minus hybrid -8.33 points. The respective paired intervals are $[-14.58, 0]$ and $[-18.75, 0]$ points. Both LongMem two-sided exact paired tests have Holm-adjusted $p = 0.4375$. The frozen witness readout supplies no demonstrated answer-quality advantage on these new samples. Its largest final LongMem deficit is temporal reasoning: 4/9 correct against dense 7/9 and hybrid 6/9. This locates a concrete requirement for the relation and time interface.

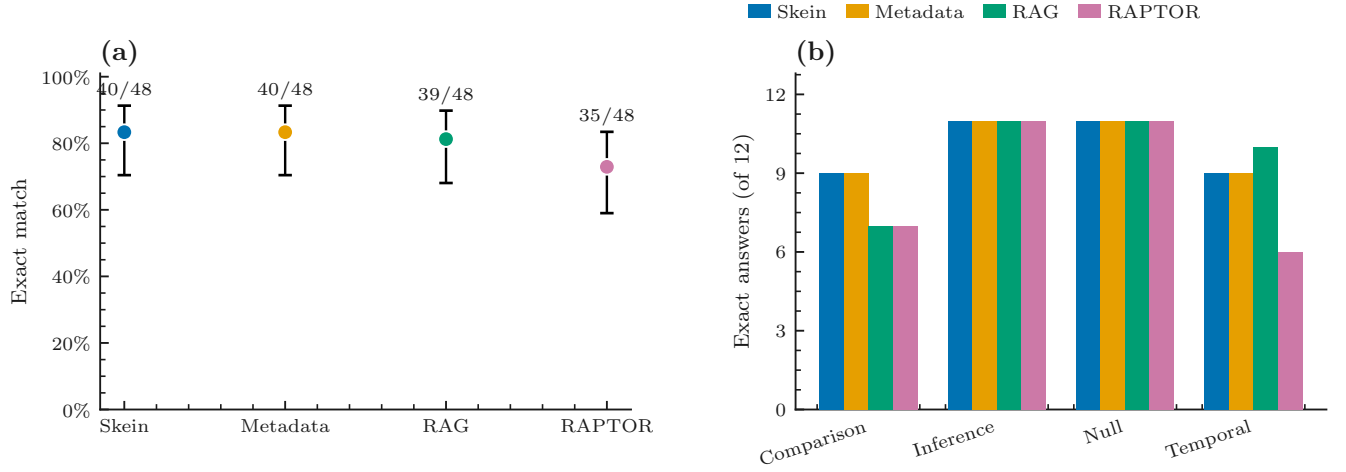


FIG. 5. The preserved 48-question news answer trial, with the original method keys retained in the figure for provenance. Source-conditioned fine retrieval answers 40 questions, flat RAG 39 and RAPTOR 35. The metadata control supplies identical contexts to the conditioned readout. Error bars are Wilson 95% intervals. This is an answer experiment; the development coverage tests in the text measure a different stage of the pipeline.

VII A MEMORY THAT CAN CHANGE ITS DESCRIPTION

The mathematical unit of recursion is the complete effective object. Its internal modes preserve observable distinctions. Its law specifies their interactions. Its decoder and incidence connect the representation to the recorded sources, while its residual identifies the need for a finer or different section. Passing these together makes repeated grading coherent.

The exact continuation experiment demonstrates the computational benefit of preserving the right shared structure: a large correlated source set can answer local enquiries through a small conditional representation. The language experiments identify a concrete next requirement for Atlas, the engine that will assemble these operations into a working memory. The observation interface must preserve the relations and aggregation operations needed by an enquiry, so that the recursive machinery is asked to conserve the right object. That requirement is testable against original evidence and the same complete-object algebra at every scale.

Appendix A: Conditional descriptions and overlapping sections

The finite state conventions are as follows. $C \in \{0,1\}^{m \times n}$ has one nonzero entry per column and no empty row. Reference masses are strictly positive. Put $\pi_c = C\pi$ and define R by Eq. (1). Since $CWC^\top = W_c$, one has $CR = I$ and $R\pi_c = \pi$. Every column of R is nonnegative and sums to one. Moreover,

$$W^{-1}RC = C^\top W_c^{-1}C \quad (\text{A1})$$

is symmetric. Thus RC is an orthogonal projector in the state inner product $\langle u, v \rangle_{W^{-1}} = u^\top W^{-1}v$. For any source test f the lifted coarse test is $C^\top R^\top f$, and

$$\|f - C^\top R^\top f\|_W^2 = \sum_a \pi_c(a) \text{Var}_\pi(f | a). \quad (\text{A2})$$

This follows by subtracting the conditional mean in each cell and summing the resulting squared deviations.

For two partitions with inherited references, $C_{02} = C_{12}C_{01}$ and

$$\begin{aligned} R_{01}R_{12} &= W_0C_{01}^\top W_1^{-1}W_1C_{12}^\top W_2^{-1} \\ &= W_0C_{02}^\top W_2^{-1} = R_{02}. \end{aligned} \quad (\text{A3})$$

An independently refitted reference changes the lift and requires a separate transport discrepancy. Reference inheritance is an assumption of the displayed composition rule.

A compatible overlapping cover is represented without multiplying source mass. Let $w_s(i) \geq 0$ and $\sum_s w_s(i) = 1$. Retain active tags (i, s) , define $Jp(i, s) = w_s(i)p(i)$, and let D_s sum tags belonging to the same source. Then $D_s J = I$. A source test lifts to $D_s^\top f$, preserving its expectation:

$$(D_s^\top f)^\top Jp = f^\top p. \quad (\text{A4})$$

Partition operations can now act on the tagged support. The membership weights are declared routing data, with their own provenance.

There is a compatible generator lift. Write $G = G_{\text{off}} - \Lambda$, with $\Lambda_{ii} = \sum_{j \neq i} G_{ji}$, and repeat each exit rate on its source tags to obtain Λ_{tag} . Define

$$G_{\text{tag}} = JG_{\text{off}}D_s - \Lambda_{\text{tag}}. \quad (\text{A5})$$

The cross-source off-diagonal entries are nonnegative, and every column sums to zero. The identities $D_s\Lambda_{\text{tag}} = \Lambda D_s$ and $\Lambda_{\text{tag}}J = J\Lambda$ imply

$$D_s G_{\text{tag}} = G D_s, \quad G_{\text{tag}} J = J G. \quad (\text{A6})$$

The lift redraws the destination tag from its declared membership after a source transition. Persistent tag identity would require a different declared operation. Contexts with incompatible observations retain their separate references and operators instead of using this compatible tagging construction.

For completeness, the classical free-energy reduction also composes. For $F(p) = \langle K \rangle_p + MD_{\text{KL}}(p \| \pi)$ with $M > 0$, fix $p_c = Cp$. The KL chain rule separates the within-cell terms. Minimising each normalised conditional law gives

$$K_{\text{eff}}(a) = -M \log \sum_{i \in a} \pi(i | a) e^{-K_i/M}. \quad (\text{A7})$$

The minimiser is proportional to $\pi(i | a) e^{-K_i/M}$. Substitution gives $F_c(p_c) = \langle K_{\text{eff}} \rangle_{p_c} + MD_{\text{KL}}(p_c \| \pi_c)$. Nested conditional sums prove consistency of two successive eliminations. This is a finite classical identity for the stated reference and scale M .

Appendix B: Effective generators and retained fluctuation spaces

For fixed C , minimise the weighted quadratic mismatch $\|(CG - HC)W^{1/2}\|_F^2$ over the coarse matrix H . Differentiation gives $(CG - HC)WC^\top = 0$, hence the unique solution

$$G_c = CGR. \quad (\text{B1})$$

For distinct coarse cells, all contributing source transitions are off-diagonal transitions of G , so $(G_c)_{ab} \geq 0$ for $a \neq b$. Furthermore, $\mathbf{1}^\top G_c = 0$ and $G_c \pi_c = CG\pi = 0$.

The corresponding positive-state defect is $\Delta = CG - G_c C$. Its two-grade identity,

$$\Delta_{02} = C_{12}\Delta_{01} + \Delta_{12}C_{01}, \quad (\text{B2})$$

follows by expanding both terms and cancelling $C_{12}G_1C_{01}$. For Markov generators, the induced 1-norm semigroup bound is $\|C e^{tG} - e^{tG_c}C\|_1 \leq t\|\Delta\|_1$. The signed retained coordinates of the main construction use a different norm and operator, derived below.

Detailed balance makes L in Eq. (2) symmetric. If $c_{ij} = G_{ij}\pi_j = c_{ji}$ for $i \neq j$, then

$$-x^\top Lx = \frac{1}{2} \sum_{i \neq j} c_{ij} \left(\frac{x_i}{\sqrt{\pi_i}} - \frac{x_j}{\sqrt{\pi_j}} \right)^2 \geq 0. \quad (\text{B3})$$

Consequently $A = U^\top L U \leq 0$ for every isometry U . The mass column in each parent is its normalised reference vector. Each internal column is orthogonal to this vector and supported inside the parent. At a later grade, the reference has nonzero entries only at the mass coordinates. The parent receives all coordinates owned by each child, including its internal register.

The constant-coordinate block of the transported operator is the whitened representation of the original conditional mass generator. If $s_c = \sqrt{\pi_c}$, it is $L_{\text{mass}} = \text{diag}(s_c)^{-1} G_c \text{diag}(s_c)$. Hidden columns affect the full signed dynamics, while positive mass routing retains its conditional meaning. A permitted change of internal coordinates fixes each mass column and rotates only the

hidden register. Under an orthogonal rotation O , the operator and decoder transform as $A \mapsto O^T A O$ and $T \mapsto O^T T$. The represented state and observable values are unchanged.

To prove Eq. (6), differentiate $e^{(t-s)A} U^T e^{sL}$ and integrate:

$$U^T e^{tL} - e^{tA} U^T = \int_0^t e^{(t-s)A} D e^{sL} ds. \quad (\text{B4})$$

Both semigroups are Euclidean contractions. Submultiplicativity yields the stated bound. Reversibility is sufficient for these particular contraction estimates. The algebraic defect identities hold more generally, with the appropriate semigroup factors retained.

Appendix C: Observable loss, spectral selection and repeated grading

Let $s = \sqrt{\pi}$ and assume $Ps = s$. For $a = W^{1/2}f$ and $e = (I - P)a$, one has $e^T s = 0$. Thus the reconstruction error is $e^T W^{-1/2}(p - \pi)$. Cauchy-Schwarz proves Eq. (4). If $p = \pi(\cdot | A)$ and $\pi(A) = b > 0$, direct substitution gives

$$\chi^2(p|\pi) = b^{-1} - 1. \quad (\text{C1})$$

The division by a conditioning mass is therefore part of the answer contract, including its small-mass behaviour.

The local Euclidean demand matrix may contain both whitened tests and law-boundary demands. Given the constant projector P_0 , typical columns are $(I - P_0)a_\alpha$ and $(I - P_0)LU_0$, with explicitly normalised family and column weights. Within a fixed accessible block, an orthonormal register B captures $\text{Tr}(B^T E E^T B)$ of the demand norm. In an eigenbasis of $E E^T$, this is a weighted sum of eigenvalues, with weights in $[0, 1]$ summing to the retained rank. The maximum is the sum of the largest eigenvalues. This proves the local spectral selection rule. It does not replace the subsequent componentwise task and full dynamical-defect checks.

A more general local correction geometry has the same direct solution. For a residual demand e , accessible chart F , readout map T , positive price matrix K and $\lambda \geq 0$, define

$$J(u; e) = \frac{1}{2} \|T(e - Fu)\|^2 + \frac{\lambda}{2} u^T K u. \quad (\text{C2})$$

On the supported quotient, put $\mathcal{H} = F^T T^T T F + \lambda K$ and $b = F^T T^T T e$. If $\mathcal{H} > 0$, completing the square gives

$$u_* = \mathcal{H}^{-1} b, \\ J(u; e) = J(0; e) - \frac{1}{2} b^T \mathcal{H}^{-1} b + \frac{1}{2} \|u - u_*\|_{\mathcal{H}}^2. \quad (\text{C3})$$

For demands b_α with weights w_α , the optimal rank- r shared corrective space is obtained from the leading eigenvectors of $\mathcal{H}^{-1/2}(\sum_\alpha w_\alpha b_\alpha b_\alpha^T) \mathcal{H}^{-1/2}$. If those orthonormal eigenvectors are v_j , the register directions in the

original correction coordinates are $\mathcal{H}^{-1/2} v_j$. They are orthonormal in the $\mathcal{H}$ metric. This normal matrix acts on the correction coordinates u , whereas the Dirichlet operator H acts on source fluctuations. A declared finite rank-cost table can then be compared with the captured spectral benefit. This identifies the assumptions under which local spectral efficiency is optimal, without equating an abstract quadratic price with hardware runtime.

For two grades let $P_1 = U_1 U_1^T$ and $P_2 = U_2 U_2^T$. The source-level observable residual splits as

$$(I - U_1 P_2 U_1^T) a \\ = (I - P_1) a + U_1 (I - P_2) U_1^T a. \quad (\text{C4})$$

The first term is orthogonal to $\text{ran}(U_1)$ and the second lies within it. Their squared norms add exactly. Expanding the definitions of D_{01} and D_{12} similarly proves Eq. (8). No child register may be removed between these two operations.

The four-source refuter uses reversible pair-reset dynamics and the test $(0, 1, 0, 1)$. A child's conditional mean is $1/2$, and one local contrast reconstructs its two test values exactly. At the next grade, retaining a suitable contrast of the full child coordinates preserves the test and its decay rate. Dropping the registers instead gives pointwise deviations $\pm 1/2$, hence reference RMS $1/2$. The two results distinguish a valid inherited-loss account from one that silently changes the retained object.

Appendix D: Eliminated modes and memory

Keep U fixed during one evolution and write $Q = I - P$, $z = U^T y$, $\eta = Qy$. Projecting $\dot{y} = Ly$ gives

$$\dot{z} = Az + U^T L Q \eta, \\ \dot{\eta} = Q L U z + Q L Q \eta. \quad (\text{D1})$$

Variation of constants on $\text{ran}(Q)$ yields

$$\eta(t) = e^{tL_Q} \eta(0) + \int_0^t e^{(t-s)L_Q} Q L U z(s) ds. \quad (\text{D2})$$

Substitution and $Q L U = D^T$ prove Eq. (9). The memory kernel is $K(t) = D e^{tL_Q} D^T$, and $\|K(t)\|_2 \leq \|D\|_2^2$. For a Laplace variable with $\text{Re } \zeta > 0$, elimination gives the retained resolvent

$$[\zeta I - A - D(\zeta I - L_Q)^{-1} D^T]^{-1}. \quad (\text{D3})$$

The middle term is the contribution of excursions through the omitted space. If $D = 0$, both cross blocks vanish by self-adjointness and the memory-free retained evolution is exact.

The positive operator $-L$ has units of inverse time when G is a kinetic generator. In the news construction, time parametrises a declared graph diffusion. The positive source state, its admissible fluctuation domain and the original evidence remain distinct from the signed

Galerkin coordinates. A moving basis would introduce connection terms involving $\dot{U}$. The present implementation changes grade by explicit discrete transports and uses fixed maps within each evolution.

Appendix E: Tensor effects, provenance and fine-passage bounds

For the real unit vectors used in Eq. (10), $\langle a_{if}, (1, q_f)/\sqrt{2} \rangle = (1 + \langle v_{if}, q_f \rangle)/2$. Let E_q be the tensor product of selected rank-one query projectors and identities on unselected fields. Then

$$\begin{aligned} \text{Tr}[\Phi(p)E_q] &= \sum_i p_i f_q(i), \\ \frac{\text{Tr}[\Omega(|i\rangle\langle i| \otimes E_q)]}{\pi_i} &= f_q(i). \end{aligned} \quad (\text{E1})$$

The averaged density $\Phi(\pi)$ suffices for aggregate expectations. Individual ranking and evidence witnesses require the source-labelled blocks. The density is a separable classical mixture of the observed field states. The tensor form records compatible correlations and does not supply a joint interpretation for incompatible operational contexts.

The exact finite categorical core stores source weights on tuples and identifies prefixes with the same conditional suffix law, in a fixed factor order. Equal normalised continuations can share the same successor representation. Induction from the terminal proves that the resulting directed acyclic graph evaluates every product-factor expectation correctly. The quotient is minimal among deterministic normalised continuation representations for that fixed order and exact equivalence relation. Source labels remain external witnesses so that unique identifiers do not destroy otherwise useful shared structure.

For the fine-passage readout, source and date conditions act on that joint source incidence. Passage rows retain exact source ownership and the ordered passage manifest. A query acts on original cached passage vectors. Eq. (12) gives

$$\begin{aligned} \langle v_j, q \rangle &= \langle \mu_b, q \rangle + \langle c_j, B_b^\top q \rangle \\ &\quad + \langle r_j, (I - B_b B_b^\top) q \rangle. \end{aligned} \quad (\text{E2})$$

Orthogonality and Cauchy–Schwarz prove Eq. (13). The maximum of valid member upper bounds bounds the best score inside a coarse source. A source mean does not supply this maximum bound.

For a selected set of k members, let ℓ_k be its smallest lower bound. If every omitted upper bound is at most $\ell_k + \varepsilon$, then no omitted exact score exceeds the smallest selected exact score by more than ε . The implementation resolves selected scores and returns them in descending order, using passage-row order on exact ties. Refining a valid interval can only increase its lower bound. This monotonicity makes a pass over all candidates whose upper bounds exceed the current threshold sufficient once

the selected scores are also resolved. Empty and smaller-than- k supports retain the same ordering and provenance rules.

An embedding observation is the declared numerical interface. A required source fact absent from that observation cannot be recovered by improving its numerical factorisation. The stored source passage and its qualifications provide the route for a richer enquiry. In the reported QA test, this final interpretation is performed by the fixed reader and measured against the dataset’s reference answers.

Appendix F: Direct matrix-product construction and conditional error

For a deterministic continuation state a and symbol x_g , let $t_g(a, x_g)$ be its transition probability and b its unique successor. Set

$$A_g[a, x_g, b] = \sqrt{t_g(a, x_g)} \quad (\text{F1})$$

on that transition and zero elsewhere. With one-dimensional root and terminal boundaries, every configuration has at most one nonzero path. Multiplication therefore gives $\psi(x_1, \dots, x_d) = \sqrt{p(x_1, \dots, x_d)}$. For a diagonal product effect O_q ,

$$\langle \psi | O_q | \psi \rangle = \sum_x p(x) \prod_g f_{q,g}(x_g). \quad (\text{F2})$$

This is the same answer as the probability-core contraction.

Continuation labels generally have overlapping suffix amplitudes. Canonical QR sweeps account for this environment before any SVD is interpreted as a Schmidt decomposition. Across one canonical cut, $\Psi = USV^\dagger$ has squared best rank- r amplitude error $\sum_{j>r} \sigma_j^2$. The variational characterisation of the leading eigenspace proves this single-cut statement. Several successive truncations are assessed through the actual final overlap, rather than being treated as one globally optimal rank constraint.

Let $\phi \neq 0$ be the final approximate state, and define

$$F = \frac{|\langle \psi | \phi \rangle|^2}{\langle \phi | \phi \rangle}, \quad \epsilon = \sqrt{1 - F}. \quad (\text{F3})$$

The difference of the two normalised pure-state density operators has, on their two-dimensional span, eigenvalues $\pm\sqrt{1 - F}$. Hence every effect $0 \leq E \leq I$ obeys

$$|\langle E \rangle_\psi - \langle E \rangle_\phi| \leq \epsilon. \quad (\text{F4})$$

For a signed observable of operator norm at most one, the corresponding bound is 2ϵ . Norms and overlaps are evaluated by tensor contraction.

For a conditional answer a/b , suppose approximate masses $\hat{a}, \hat{b}$ have errors at most ϵ_a, ϵ_b and $\hat{b} > \epsilon_b$. Subtracting the fractions gives

$$\left| \frac{a}{b} - \frac{\hat{a}}{\hat{b}} \right| \leq \frac{\epsilon_a + |\hat{a}/\hat{b}| \epsilon_b}{\hat{b} - \epsilon_b}. \quad (\text{F5})$$

If the denominator interval reaches zero, the requested conditional certificate requires refinement. Filtering physical tensor legs by the query changes the environments and can change the useful local ranks. An absolute global truncation error must still be compared with the conditioning mass.

The fluctuation $W^{-1/2}(p - \pi)$ and amplitude $\sqrt{p}$ are related nonlinearly. Near π , $\sqrt{p} = \sqrt{\pi} + \frac{1}{2}W^{-1/2}(p - \pi) + O((p - \pi)^2)$ componentwise on fixed positive support. This local relation does not identify their global spectra or transfer a rank bound to the feature density in Eq. (10).

Appendix G: Original GPU representation and numerical conventions

For a reversible sparse source graph, orient every undirected edge $e = (i, j)$ arbitrarily and define $b_e = \sqrt{c_{ij}}(e_i/\sqrt{\pi_i} - e_j/\sqrt{\pi_j})$. Equation (B3) gives

$$L = - \sum_e b_e b_e^\top. \quad (\text{G1})$$

The implementation stores the sparse symmetric operator and block-local isometry coefficients. It never requires a dense global composite U . Parent labels are extended through the coordinate-owner array, so every mass and hidden coordinate enters its parent's block.

Sparse transport groups source contributions by parent-pair key before accumulating local outer products. For local ranks r_a, r_b , a contribution joining those blocks has work proportional to $r_a r_b$. Stored isometry coefficients scale with the sum of local input widths times retained ranks, and operator storage with the retained block-pair sizes. Index grouping, padded slots and duplicate contributions are counted separately. Fixed ranks and bounded sparse connectivity give the measured linear family. Dense connectivity or increasing local ranks changes that bound.

For a block representation of D , define $M_{ab} = \|D_{ab}\|_F$. Applying the triangle inequality to block vector norms gives

$$\|D\|_2 \leq \|M\|_2 \leq \sqrt{\|M\|_1 \|M\|_\infty}. \quad (\text{G2})$$

Together with Eq. (8), this gives a computable conservative composed bound. A source-operator norm bound can cap the sum because $\|U^\top LQ\|_2 \leq \|L\|_2$. The task account stores inherited squared residuals and new orthogonal losses by task identity.

Source-local passage frames use padded, bounded blocks and batched Gram eigendecompositions, followed by orthogonalisation. Coordinate and residual calculations are chunked. The preflight includes input, output, backend workspace and retained storage. A separate adapter preflight covers simultaneous source-index loading and shared passage observations. The direct route materialises only the incidence and source CSR needed

by its query family, leaving unused banked frame arrays off the GPU.

The hardware was an NVIDIA RTX A2000 Laptop GPU with 4 GB memory. The fixed runtime used CPython 3.14.7, PyTorch 2.11 with CUDA 12.8, Triton 3.6 and the existing BGE ONNX encoder. The encoder and passage vectors are shared across methods. The ONNX representation is an encoder implementation choice. The hierarchy, contractions and residual calculations use the GPU tensor implementation.

Exact algebraic identities in this paper concern real arithmetic. The retained operator and verification calculations use float64. Source/query embeddings use the shared float32 interface, and the passage rank calculations resolve the declared scores in float64. The fine readout uses additive tolerance 10^{-5} and an empirical numerical allowance 10^{-6} . Isometry, gauge, composed-operator, ordered-answer and residual enclosures are checked against independent calculations. These are measured floating-point checks, without a formal interval-arithmetic enclosure of every backend operation.

The banked exact-continuation comparison uses a weighted even-parity law with 2048 rows, 12 factors and continuation width 14. For 256 queries, ordinary quotient contraction takes 5.027 ms, a fused Triton kernel 0.812 ms, sparse MPS contraction 1.032 ms and direct rows 0.794 ms. The fused kernel improves its unfused implementation by a factor of 6.19, while direct evaluation remains slightly faster for this instance. The plan occupies 2952 bytes and takes 6.47 ms to build. Small-batch direct wins and cold timings remain in the saved receipt.

Other controls preserve the negative cases that delimit the construction. A rare Bernoulli event of mass 0.001 has reference squared mean error 0.000999 and rare-world squared error 0.998001. Insufficient MPS rank fails the final overlap certificate. Older frame benchmarks that pre-computed all raw scores cannot use logical leaf-opening counts as their executed savings. A previously reported 500 000-row timing lacks a recoverable raw result record and is excluded from the verified measurements.

Appendix H: The news instantiation

The finite construction is instantiated with four article observations: object, mechanism, conditions and consequences. Each is a source-only model description of at most 30 words. The request asks for specific entities, relations, qualifications and outcomes, with an empty field when none is stated, and one to three supporting quotations. Quotations are checked against the original article. The cards and their 384-dimensional BGE vectors are frozen inputs, so reproduction of the numerical hierarchy uses the archived observations rather than resampling those descriptions.

Each field has its own source context with $\pi_i = 1/609$. Form the directed 16-nearest-neighbour graph from the banked unit vectors using $h(t) = ((1 + t)/2)^2$ of cosine

similarity. Its symmetric edge weight a_{ij} is the larger of the two directed weights. All edges incident to a missing-field vertex are then removed. Define

$$r_j = \sum_{i \neq j} \frac{a_{ij}}{\pi_j}, \quad \kappa = \max(1, \max_j r_j),$$

$$G_{ij} = \frac{a_{ij}}{\kappa \pi_j} \quad (i \neq j), \quad G_{jj} = -\frac{r_j}{\kappa}. \quad (\text{H1})$$

The conductances $G_{ij}\pi_j = a_{ij}/\kappa$ are symmetric and the largest exit rate is at most one. This fixes the time scale of the reported diffusion and closure bounds.

The task ensemble gives half its weight to the known source directions and half to unit differences from each source’s nearest neighbour. Differences with norm at most 10^{-7} are omitted, without selecting a replacement neighbour. Diagonalising the weighted direction metric gives an equivalent probe basis. Eigenvalues above $10^{-7} \max(1, \lambda_{\max})$ are retained, then normalised to sum to one. The source test columns are the field observations projected onto these directions and multiplied by $\sqrt{\pi_i}$. This field-space calculation specifies the tasks used to select the distinct source-state registers.

For a proposed parent count k , the cut proposal uses the leading k eigenvectors of the whitened mass generator, scaled by $e^{t\lambda_j}$. GPU Lloyd iterations cluster those diffusion coordinates. Initial centres are selected by farthest-point seeding from the first row, and 24 iterations are performed. The first cut is reused from the frozen source graph. Later reference cuts are proposed from the transported mass block. The compact implementation replays those exact cut labels when comparing with the dense oracle. The procedure supplies reproducible proposals, while task and law acceptance determine whether a grade is admitted.

At each proposed grade, the local demand matrix combines the task columns with $0.1(I - P_0)LU_0 \text{diag}(\sqrt{\pi_{\text{parent}}})$. Each family receives weight $1/2$. The componentwise task check includes the orthogonally inherited squared loss. The compact law check uses the composed norm bound times the declared horizon. These are the parameters of the corrected retained-register construction, rather than the earlier mean-only acceptance rule.

The passage readout has a separate, explicit query grammar. It matches case-folded source names at word boundaries, including the prefix before a vertical bar and names with an initial *the* removed. Overlapping matches prefer the longer name. Clauses separated by *while* or *whereas* supply the corresponding facet text. A recognised source is bound to the nearest recognised date following its mention before the next source, or to the last shared leading date. The frozen case-insensitive date pattern recognises full month names, ordinal or ordinary days, and a four-digit year beginning with 20. Unmatched dates remain unbound. With no source mention, the complete question is one unconstrained facet. These rules are shared with the metadata control.

The facet text is encoded in the shared BGE query mode. Exact joint source/date incidence selects eligible source IDs, whose original passages are reached through a source CSR. Frames are built within source-local blocks in passage order. The direct and frame paths rank the same normalised original-passage inner product, with a stable passage-row tie rule. The final direct choice follows the fixed 32-facet development timing comparison. Thus the observed 48-question context identity with the metadata control is an explicit execution check, rather than evidence that the four article fields contributed additional answer information.

Appendix I: Dataset, protocol and complete result record

The public MultiHop-RAG corpus contains 609 articles and 2556 questions. Passages contain at most 100 `c1100k_base` tokens and do not overlap. The query identifier is the SHA256 hash of its UTF-8 text. Within each of the four question types, the first four identifiers in sorted order form development. Positions 4–15 form the first trial, and positions 16–27 form the independent final trial. Each test has 12 questions per type and 48 in total. Results are reported separately.

All arms use the same fixed Sol reader and low reasoning effort. The supplied context is the complete canonical JSON passage list, including identifiers, sources and dates, limited to 2000 tokens and 20 objects. The reader returns a structured answer and evidence IDs. The fixed scorer measures normalised exact match, token F1, source ancestry recall at 8 and 20 objects, and literal fact coverage. Source ancestry is a reachability diagnostic, while literal coverage tests the exact supplied text. Neither replaces answer accuracy.

Flat RAG retrieves by BGE cosine score, considers 40 candidates and uses MMR with coefficient 0.7. The metadata control scores all original passages under the same explicit source/date grammar used by the proposed readout. Final *Parallax* ranks 20 passages per facet and packs them in the same round-robin order across facets. The reference RAPTOR implementation follows its independent global/local UMAP and GMM construction, with embedding dimension 10, BIC selection from 1 to 49 components, soft membership above 0.1, 3500-token cluster limit, 100-token summaries, maximum five levels and seed 224. Its complete tree has 14 359, 1410, 271, 55, 11 nodes by level, or 16 106 total. RAPTOR uses its collapsed-tree retrieval path.

Preprocessing costs are preserved separately from retrieval. The 609 source cards require 26 batched requests and 852.4 s elapsed. The RAPTOR hierarchy contains 1747 summary jobs in 61 batched requests. Its recorded levelwise summary elapsed times sum to 1488.9 s, CPU preflight times to 172.9 s, and embedding times to 8.40 s. The final fine passage build requires no additional source-summary request. Its 0.472 s construction reuses

TABLE II. Declared parameters of the news construction. A register rank counts hidden coordinates in addition to the parent’s mass coordinate. Only grades that reduce the number of objects and pass the specified tests are accepted.

Quantity	Instantiation
Source reference	Uniform on 609 source IDs, inherited at every grade
Observed fields	Object, mechanism, conditions, consequences
Graph	Symmetric union of 16 nearest neighbours, maximum exit rate one
Proposed parent counts	24, 6, 1, 1
Diffusion times	1, 2, 4, 8
Minimum non-singleton size	2, with missing observations isolated
Maximum hidden rank per parent	8
Task and law-family weights	1/2, 1/2
Law-demand scale and prediction horizon	0.1, 0.1
Componentwise task RMS thresholds	0.15, 0.30, 0.50, 0.70
Composed horizon-weighted law thresholds	0.20, 0.30, 0.40, 0.50
Reported near-tie tolerance	10^{-7}
Fine passage block width and tested ranks	32 and 4, 8, 16
Final fine candidates per facet	20, with additive tolerance 10^{-5}

TABLE III. Both fixed trials, each with 48 questions. The article-gated and corrected results use disjoint questions and are not pooled. Parentheses give correct answers by comparison, inference, null and temporal type, respectively, each out of 12.

Method	First trial	Final trial
<i>Parallax</i>	33 (5, 10, 12, 6)	40 (9, 11, 11, 9)
Metadata	40 (10, 11, 11, 8)	40 (9, 11, 11, 9)
Flat RAG	33 (8, 10, 11, 4)	39 (7, 11, 11, 10)
RAPTOR	35 (7, 11, 11, 6)	35 (7, 11, 11, 6)

the banked passage vectors. The compact retained-state run records 3.2615 s of grade construction, 0.1225 s of separately timed comparison, and 3.6731 s for the enclosing cycle including setup and receipt handling.

In the first trial, the proposed readout tied RAG and scored two below RAPTOR. The adjusted one-sided probabilities were both one. Its hard three-article gate was then removed using the fine-source sufficiency argument already recorded before that trial and an independent audit of the original development questions. No question from the first trial was included in the final test. A registration-format refusal during final preparation is retained with its original input tag. The metadata-only correction left the selected questions, method, reader and scores unchanged and preceded all reader predictions.

The final exact-match differences are 1/48 against RAG and 5/48 against RAPTOR. The raw paired one-sided probabilities are 0.5 and 0.03125. Holm correction over the two planned comparisons gives 0.5 and 0.0625. Descriptive paired 95% bootstrap intervals for the accuracy differences are $[-6.25, 10.42]$ and $[2.08, 18.75]$ percentage points, respectively. The bootstrap uses 10 000

stratified draws with PCG64 seed 20260919 and persists its actual index array. The descriptive interval and the multiplicity-adjusted decision have different roles.

Final token F1 is 0.8472 for *Parallax* and metadata, 0.8125 for RAG, and 0.7431 for RAPTOR. Source ancestry recall at 8 is 0.7685, 0.7685, 0.6458, 0.7546 in the same order, and at 20 it is 0.8241, 0.8241, 0.7639, 0.8009. These source metrics average the 36 questions with reference evidence. The 12 null questions have no such denominator. Literal coverage is 0.2755, 0.2755, 0.1597, 0.1250. All reference sources map to the corpus, and every returned citation ID belongs to its supplied context.

The final run has 192 successful answer records, zero failures, zero retries and zero observed reader tools. Per-request locking gives 143 actual reader executions and 49 cache-hit records. All 48 proposed contexts, request hashes and answers equal the metadata control. One further identical request involves RAPTOR. Actual reader times sum to 730.8 s across concurrent requests, which is not elapsed wall time. The first trial made 192 actual calls because eleven duplicate request pairs raced concurrently. That cost is preserved.

Appendix J: Reproducibility and evidence custody

The source repository contains the complete code, queries, stored contexts, responses, numerical receipts and independent reviews. Inputs are hashed before prediction, the complete prediction matrix is sealed and published before scoring, and the score manifest binds the analysis and bootstrap indices back to those two records. Verification requires no new reader call.

The final input publication is the lightweight Git tag `skein-fine-inputs-20260920-v3a`, at commit `1511933b60e2`. The prediction publica-

tion is `skein-fine-predictions-20260920-v3a`, at `a277d2ec93d9`. The score record is committed at `ce6eb876`. These records are in the [Laplace repository](#), under `synthetics papers/corpus-as-a-state/reconstruction`. The input protocol self-hash begins `85fbb90099d0`, and the run lock self-hash begins `3d5095fe3415`. Full SHA256 values and the hashes of literal files are stored in the manifests, with those two hash meanings distinguished.

The principal numerical records are `section_compact_news_parity_v5_frozen_core.json`, `section_block_operator_scaling_v4_capacity_accounted.json`, and `passage_residual_probe_v1.json` in the reviews directory. The fine source index begins `e21c27b2333f`, the compact news state `ad0181878d56`, and the original native query index `7497ce1a2e6f`. Verified archives retain those representations and the costly source-card and RAPTOR construction results. Figure generators carry source hashes, exact JSON locators and reproduction commands in their provenance record.

The mathematical record includes the original source-reading manifest, the retained-register derivation, the fine-source correction contract and independent reviews of the operator, tensor and memory connections. Implementation review includes numerical refuters for rare conditioning, hidden coupling, gauge transport, omitted modes, candidate ordering and capacity refusal. Protocol review checks source identity, schedule completeness, reader isolation, publication order and score integrity.

Hashes establish identity and detectable change. The published records make the declared sequence inspectable, while command outputs, independent calculations and reviews supply the evidence for the computations themselves. The public dataset and remotely served reader also leave pretraining exposure and served model weights outside the local numerical attestation.

Appendix K: Primitive integer continuation

Let state s have primitive integer suffix counts $c_s(x)$, denominator T_s and greatest common divisor one on its supported counts. A terminal has $T = 1$. An outgoing edge of reduced probability n/d to state b gives branch probabilities $nc_b(x)/(dT_b)$. Since the child counts are primitive, the common reduced denominator on this branch is

$$B_{sa} = dT_b / \gcd(n, T_b). \quad (\text{K1})$$

Distinct symbol branches have disjoint configurations. Their least common multiple is therefore the parent primitive denominator:

$$T_s = \text{lcm}_a B_{sa}, \quad K_{sa} = \frac{T_s}{B_{sa}} \frac{n}{\gcd(n, T_b)}. \quad (\text{K2})$$

Induction gives the primitive parent count $K_{sa}c_b(x)$ on each branch, and normalisation gives $\sum_a K_{sa}T_b = T_s$.

Multiplying by Boolean query factors proves Eq. (14). If the original empirical total is W , the unnormalised answer is $m_{\text{root}}W/T_{\text{root}}$. The denominator divides W . Numerator and denominator contractions yield conditional answers, with a zero denominator reported as undefined.

The implementation reduces before multiplication and checks every least common multiple, multiplier and normalisation sum for overflow. The proved bound $m_s \leq T_s$ then licenses the selected integer message width. This argument applies to Boolean effects on an integer empirical law. Real vector effects and amplitude truncations use the separate contracts derived above.

Appendix L: Extension evaluation and source custody

The continuation input tag is `reconstruction-coordinate-inputs-20260920-v1`, published before its 524-cell matrix. The record `extension/results/query_locality_matrix_v1.json` contains every measured cell and capacity refusal. The vector-locality experiment is preserved separately in `extension/results/locality_matrix_v1.json`.

LongMemEval uses the original cleaned S corpus and a fixed hash-selected 24-case development / 48-case final split. Opaque identifiers cross the model boundary. Source projection admits conversation text and metadata; answer text, answerability, question type and positive-support labels belong to the evaluator. QASPER uses a 32-question official training development bank and 256 fixed questions from the official development split. Its paragraph/caption evidence score chooses one human annotation and preserves unmapped requirements in that annotation’s denominator.

Extension budgets count the entire canonical JSON context in `o200k_base`, including identifiers and metadata. LongMem’s atomic evidence unit is a primary turn with its immediate predecessor. An item supplied as a predecessor is not thereby an accepted primary. All arms use the same one-step grouping and exact whole-context token check. The sealed records preserve the groups actually accepted by the packer.

Development variants and failed attempts remain separate immutable records. A four-case evaluator repair corrected one accidental construction of an unselected gold record before a failed lookup. No value from that record was scored, exposed in the return or used to select the method. The repaired iterator tests the locked ID before reading evaluator fields. This access correction is preserved with the original failure and the independent review.

The public method name in this revision is Parallax. Existing files and method keys retain their original Skein labels so their published identities remain stable. Figure provenance records bind exact input hashes and JSON locations. Hashes permit identity checks and expose later changes; independent recomputations and the stored out-

puts provide the numerical evidence.

The final reader contexts are published at `reconstruction-reader-contexts-20260920-v1`; the complete predictions at `reconstruction-reader-predictions-20260920-v1`, before answer scoring. All arms use the same Sol low-effort reader, no tools, exact 4000-token context budget and response schema. Exact requests are shared across arms with an append-only attempt ledger. QASPER

retains its native v0.3 F1 evaluation. LongMem uses the pinned official per-type rubric with a Sol judge and is labelled an adapted score. The two planned paired comparisons use 10,000 deterministic bootstrap resamples and two-sided exact McNemar tests for binary outcomes, with Holm adjustment within each dataset. Failures remain in the full denominators. The v2 release manifest connects these records to the unchanged v1 release and both revised deliverables.

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